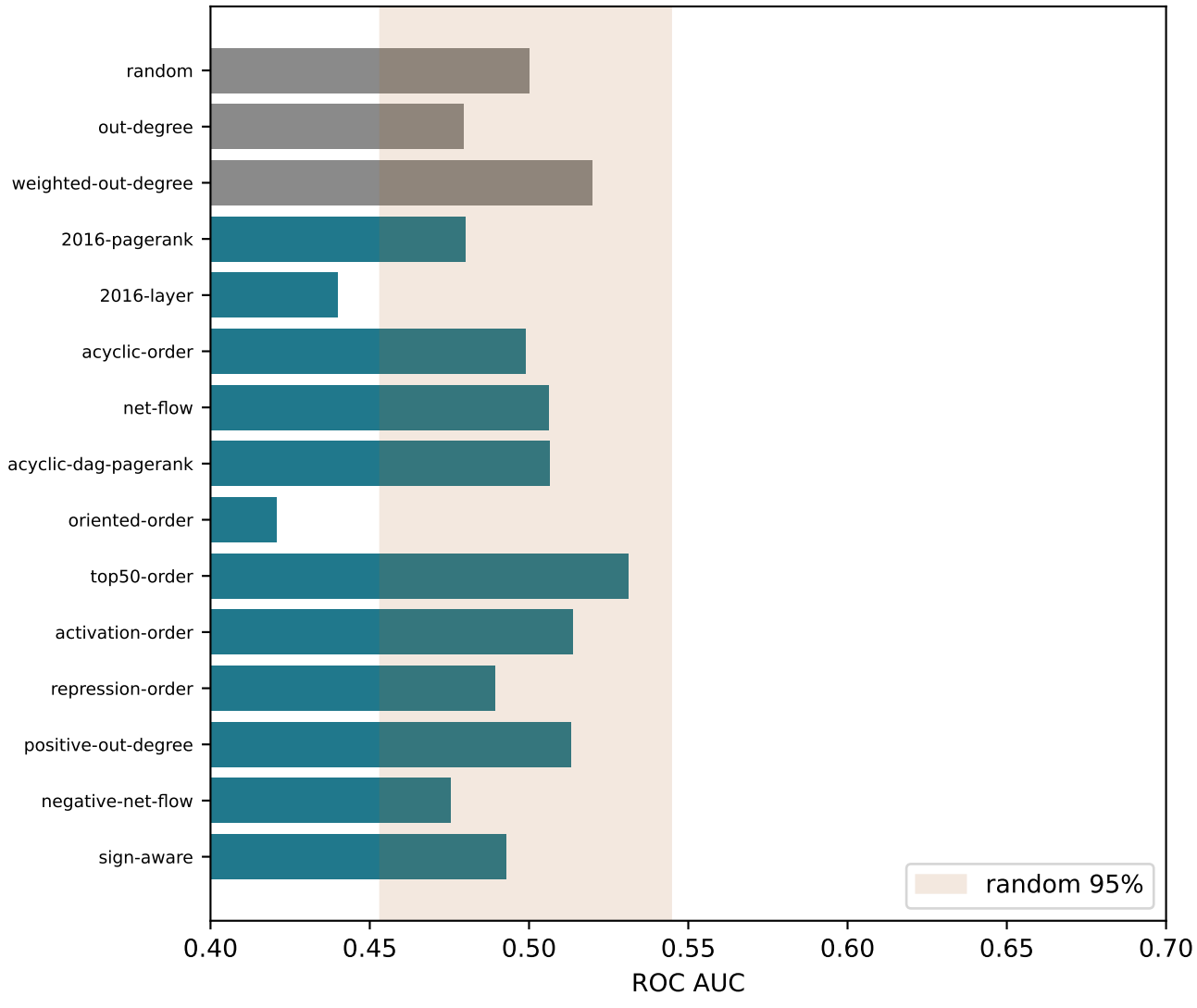
Regulator ranking against reference 'regulators'

2026-11-07 · 740 genes ranked, 200 in regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	27.0 (19–35)	0.500 (0.453–0.545)				
out-degree	24	0.479	0.802	0.803	-0.040 (-0.049, -0.032)	< 0.001
weighted-out-degree	26	0.520	0.639	0.207	+0.000 (+0.000, +0.000)	
2016-pagerank	28	0.480	0.448	0.797	-0.040 (-0.078, -0.005)	0.032
2016-layer	61	0.440	< 0.001	0.992	-0.080 (-0.118, -0.043)	< 0.001
acyclic-order	28	0.499	0.448	0.523	-0.021 (-0.046, +0.003)	0.082
net-flow	26	0.506	0.639	0.409	-0.013 (-0.025, -0.003)	0.018
acyclic-dag-pagerank	30	0.506	0.269	0.409	-0.013 (-0.030, +0.003)	0.126
oriented-order	19	0.421	0.975	1.000	-0.099 (-0.130, -0.070)	< 0.001
top50-order	29	0.531	0.357	0.094	+0.011 (-0.029, +0.049)	0.608
activation-order	32	0.514	0.138	0.292	-0.006 (-0.036, +0.024)	0.686
repression-order	26	0.489	0.639	0.665	-0.030 (-0.063, +0.001)	0.058
positive-out-degree †	23	0.513	0.862	0.302	-0.006 (-0.021, +0.008)	0.358
negative-net-flow †	21	0.475	0.946	0.847	-0.044 (-0.064, -0.025)	< 0.001
sign-aware †	26	0.493	0.639	0.611	-0.027 (-0.037, -0.016)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.

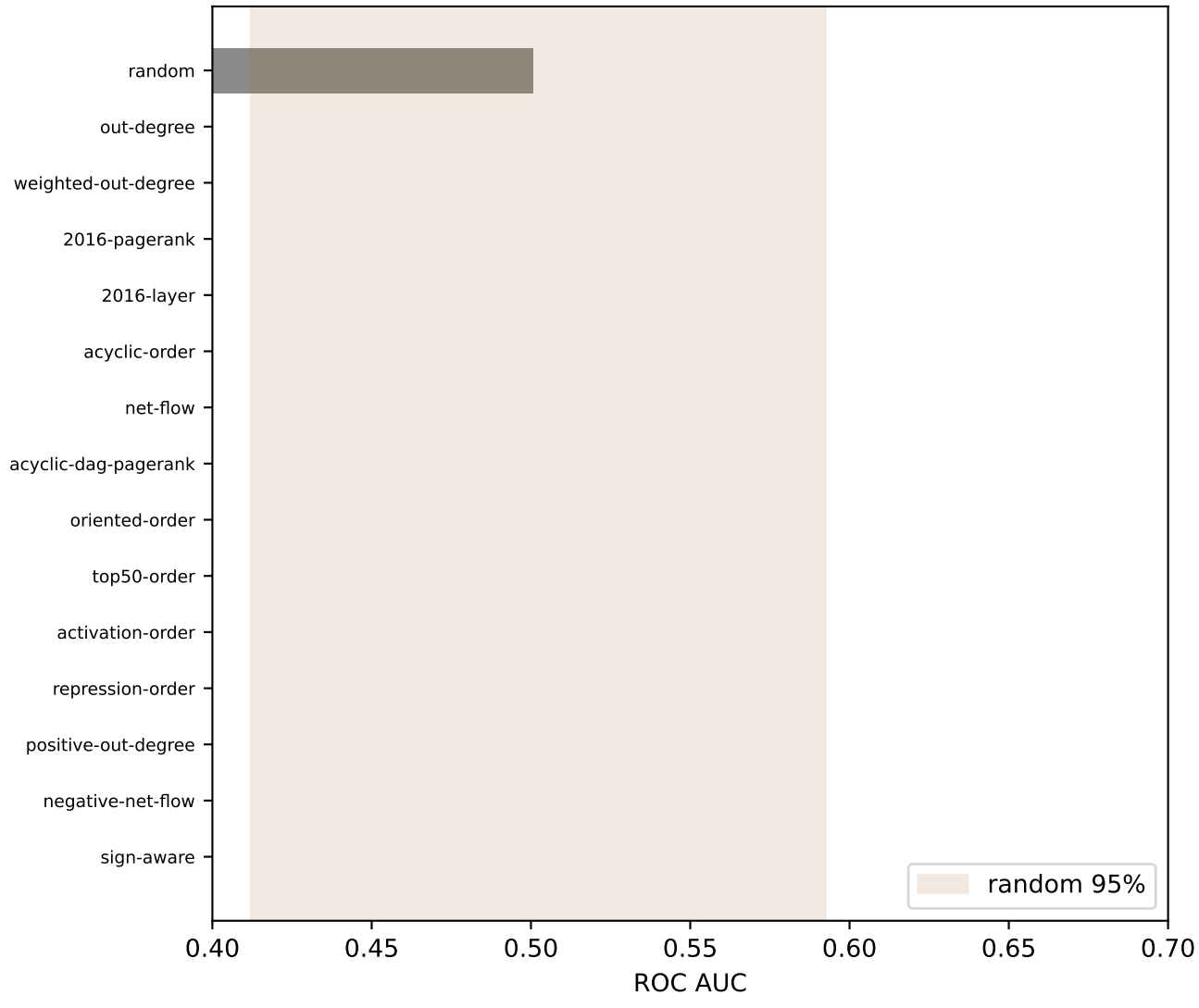


Regulator ranking against reference 'top-layer'

2026-11-07 · 740 genes ranked, 41 in top-regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	5.5 (2–10)	0.501 (0.412–0.592)				
out-degree	2	0.242	0.977	1.000	-0.036 (-0.052, -0.021)	< 0.001
weighted-out-degree	2	0.278	0.977	1.000	+0.000 (+0.000, +0.000)	
2016-pagerank	2	0.334	0.977	1.000	+0.055 (-0.030, +0.136)	0.218
2016-layer	5	0.277	0.674	1.000	-0.001 (-0.059, +0.059)	0.966
acyclic-order	2	0.269	0.977	1.000	-0.009 (-0.064, +0.041)	0.722
net-flow	2	0.281	0.977	1.000	+0.003 (-0.026, +0.035)	0.892
acyclic-dag-pagerank	2	0.268	0.977	1.000	-0.010 (-0.042, +0.020)	0.490
oriented-order	0	0.211	1.000	1.000	-0.068 (-0.117, -0.016)	0.016
top50-order	1	0.314	0.996	1.000	+0.036 (-0.022, +0.091)	0.200
activation-order	2	0.277	0.977	1.000	-0.001 (-0.050, +0.044)	0.926
repression-order	2	0.281	0.977	1.000	+0.003 (-0.062, +0.071)	0.880
positive-out-degree †	1	0.255	0.996	1.000	-0.023 (-0.039, -0.008)	0.002
negative-net-flow †	2	0.264	0.977	1.000	-0.014 (-0.057, +0.028)	0.456
sign-aware †	2	0.249	0.977	1.000	-0.030 (-0.055, -0.005)	0.020

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

true

Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	3,762	0.714 (25.40×)	29.88×	18.92×	1.00 (0.58)
resolve_reciprocals → drop_sinks → prune_triangles	3,712	0.087 (19.03×)	24.91×	16.82×	1.00 (0.54)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	2,977	0.032 (12.04×)	12.97×	11.78×	1.00 (0.64)
absolute	3,762	0.714 (25.40×)			
absolute → minimum_feedback_order	3,762	0.630 (24.70×)			
orient_reciprocals ratio=2.0 → absolute	3,762	0.468 (21.27×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	3,762	0.468 (21.27×)			
top_outgoing k=50 → absolute	3,762	0.607 (33.21×)			
top_outgoing k=50 → absolute → minimum_feedback_order	3,762	0.523 (32.50×)			
keep_sign sign=1 → absolute	3,762	0.497 (29.88×)			
keep_sign sign=1 → absolute → minimum_feedback_order	3,762	0.436 (28.88×)			
keep_sign sign=-1 → absolute	3,762	0.217 (18.92×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	3,762	0.182 (17.71×)			

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 45,399 edges (20,327 negative), largest strong component 725.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 45,399 edges (20,327 negative), largest strong component 725.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 7,044 edges (5,229 negative), largest strong component 36.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 7,044 edges (5,229 negative), largest strong component 36; DAG keeps 3,388 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 45,399 edges (0 negative), largest strong component 725; DAG keeps 40,183 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 45,399 edges (0 negative), largest strong component 725.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 45,399 edges (0 negative), largest strong component 725; DAG keeps 40,183 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 35,516 edges (0 negative), largest strong component 1; DAG keeps 35,516 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 27,372 edges (0 negative), largest strong component 725; DAG keeps 23,839 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 25,072 edges (0 negative), largest strong component 636; DAG keeps 22,138 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 20,327 edges (0 negative), largest strong component 610; DAG keeps 18,072 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 45,399 edges (20,327 negative), largest strong component 725.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 45,399 edges (20,327 negative), largest strong component 725.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 45,399 edges (20,327 negative), largest strong component 725.